LITERATURE REVIEW

Conformal Prediction Under Covariate Shift

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ANALYTICAL ABSTRACT

The paper asks whether conformal prediction, which delivers finite-sample, distribution-free prediction intervals under exchangeability, can be made to work when the test covariates are drawn from a different distribution than the training covariates while the conditional law of the response is unchanged. The authors show that if the likelihood ratio between test and training covariate laws is known, each nonconformity score can be reweighted by a probability proportional to that ratio, and the interval is read from the quantile of the weighted score distribution with the test point’s mass placed at infinity. Coverage of at least one minus alpha is proved through a new notion, weighted exchangeability, and a weighted quantile lemma. Weighted split conformal prediction is described for computational efficiency, and the ratio is estimated by probabilistic classification. On the UCI airfoil data with an exponential tilt, ordinary split conformal covers about 82 percent against a 90 percent target, whereas weighted versions return to about 91 percent at the price of wider and more dispersed intervals through a reduced effective sample size. The main limitation is that the guarantee rests on a known, or very accurately estimated, ratio and on a covariate shift assumption that cannot be verified from data.

1 Introduction

1.1 Paper identification and classification

The paper is a methodological contribution in theoretical statistics. Its field is nonparametric predictive inference, and its subfield is conformal prediction under distribution shift. The version read in full is the arXiv version (identifier 1904.06019), consulted as a complete text rendering that includes the abstract, all three technical sections, the discussion, the footnoted statements, the acknowledgements and the reference list. The paper has no appendix in the version read. The four figures are described in the text and captions, but the image content itself could not be inspected in this review; every number quoted from the experiment is therefore taken from the running text and labelled as such. A DOI is not displayed on the version read, and none is asserted here.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

It asks how to build a prediction band whose coverage probability is guaranteed in finite samples, with no assumption on the data distribution, when the test point is not exchangeable with the training sample because its covariates come from a shifted distribution.

WHY DOES THE QUESTION MATTER?

Almost every deployed predictive model is applied to a population that differs from its training population: a different period, a different customer base, a different market regime. Standard conformal prediction is the main tool that gives honest, model-agnostic uncertainty statements, but its guarantee is destroyed by exactly this situation. A method that restores the guarantee under a clearly stated shift model has direct practical value, and it also clarifies what exchangeability is actually used for in the proof.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

The conformal framework of Vovk, Gammerman and Shafer (2005) gives coverage under exchangeability, and Lei et al. (2018) established the upper bound and the regression implementation. Covariate shift was studied for point estimation, model selection and cross-validation through importance weighting (Shimodaira, 2000; Sugiyama et al., 2007). What was missing is a correction of distribution-free prediction intervals themselves, which the authors state is, to their knowledge, new.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes weighted conformal prediction. The investigation is mainly mathematical: define weighted exchangeability, prove a weighted quantile lemma, and deduce coverage. The empirical

part is a repeated-splitting experiment on the airfoil data. The authors find that the weighted band restores nominal coverage on average under covariate shift, that estimated weights from logistic regression or random forests work nearly as well as the oracle weights in that example, and that the price is dispersion caused by a smaller effective sample size.

2 Research Problem and Hypothesis

2.1 Research question

PAPER Given training pairs (X_i, Y_i), $i = 1, \dots, n$, drawn independently from $P = P_X \times P_{Y|X}$, and a test pair drawn independently from $\tilde{P} = \tilde{P}_X \times P_{Y|X}$ with the same conditional law, can a band $\hat{C}_n$ be constructed from the training data alone so that $\mathbb{P}\{Y_{n+1} \in \hat{C}_n(X_{n+1})\} \geq 1 - \alpha$ in finite samples, for every P and every absolutely continuous $\tilde{P}_X$, when $d\tilde{P}_X / dP_X$ is known?

2.2 Hypotheses

- If the nonconformity scores are weighted by the normalised likelihood ratio of test to training covariates, with the test point carrying its own weight at infinity, the resulting quantile band has coverage of at least $1 - \alpha$ under covariate shift.
- The guarantee extends to any data satisfying a weighted form of exchangeability, which includes independent but non-identically distributed draws.
- When the ratio is unknown, a probabilistic classifier trained to separate training from unlabeled test covariates gives weights that retain approximately nominal coverage.

2.3 Existing literature and research gap

PAPER The authors position the work between conformal inference under exchangeability and the covariate shift literature, which focused on correcting estimators and model evaluation rather than prediction intervals.

CRITIQUE The gap is real and precisely defined. The gap between a known and an estimated ratio is, however, left largely empirical: the only evidence for estimated weights is one low-dimensional example in which logistic regression is well specified by construction, since the tilt in Equation (10) is exactly log-linear.

2.4 Claimed contribution

CRITIQUE Three contributions are claimed: the weighted construction, the notion of weighted exchangeability with a general coverage theorem, and the applications sketched in the discussion. The first two are demonstrated rigorously and are the lasting content. The third is illustrated by examples without theorems, and the claim that estimated weights give correct coverage is supported only by simulation in one setting. The paper does not prove any coverage guarantee for estimated weights.

3 Theoretical Framework and Mechanism

3.1 Standard conformal prediction and the quantile lemma

PAPER Write $\text{Quantile}(\beta; F) = \inf\{z : \mathbb{P}\{Z \le z\} \ge \beta\}$ for a distribution F on the extended line $\mathbb{R} \cup \{\infty\}$, and $\text{Quantile}(\beta; v_{1:n})$ for the quantile of the empirical law $\frac{1}{n}\sum_i \delta_{v_i}$. The key fact behind all conformal methods is the following. **Lemma 1.** If $V_1, \dots, V_{n+1}$ are exchangeable, then for any $\beta \in (0,1)$

$$\mathbb{P}\{V_{n+1} \le \text{Quantile}(\beta; V_{1:n} \cup \{\infty\})\} \;\ge\; \beta, \tag{1}$$

and if ties have probability zero, the probability is at most $\beta + 1/(n+1)$. The proof has two steps. Replacing every support point above the β-quantile by any larger value leaves the quantile unchanged, so $V_{n+1} \le \text{Quantile}(\beta; V_{1:n} \cup \{\infty\})$ holds if and only if $V_{n+1} \le \text{Quantile}(\beta; V_{1:(n+1)})$, which in turn holds exactly when V_{n+1} is among the $\lceil \beta(n+1) \rceil$ smallest of the $n+1$ values. Exchangeability makes the rank of V_{n+1} uniform, so the probability is at least $\lceil \beta(n+1) \rceil/(n+1) \ge \beta$, and equals $\lceil \beta(n+1) \rceil/(n+1) \le \beta + 1/(n+1)$ without ties.

PAPER Let $Z_i = (X_i, Y_i)$, $Z_{-i} = Z_{1:n} \;\; \{Z_i\}$ and let $\mathcal{S}$ be a score function, for example the absolute residual of a regression fitted by an algorithm $\mathcal{A}$:

$$\mathcal{S}((x,y), Z) = |y - \hat{\mu}(x)|. \tag{2}$$

For each candidate value (x,y) the scores are computed on the augmented sample,

$$V_i(x,y) = \mathcal{S}\big(Z_i,\, Z_{-i} \cup \{(x,y)\}\big), \quad i = 1, \dots, n, \qquad V_{n+1}(x,y) = \mathcal{S}\big((x,y), Z_{1:n}\big), \tag{3}$$

and the conformal band collects every y that does not look atypical:

$$\widehat{C}_n(x) = \Big\{ y : \; V_{n+1}(x,y) \le \text{Quantile}\big(1-\alpha;\; V_{1:n}(x,y) \cup \{\infty\}\big) \Big\}. \tag{4}$$

$\mathcal{S}$	Nonconformity score: large when the point is atypical relative to the rest of the sample.
$V_i(x,y)$	Score of observation i in the sample augmented by the hypothesised test point; symmetric in the sample, which is what transfers exchangeability from the data to the scores.

$\{\infty\}$ The mass $1/(n+1)$ allotted to the unseen test score; it is the device that converts the n observed scores into an exactly valid $(n+1)$-sample rank statement.

$1-\alpha$ Target coverage level.

Equation (4) says: include y whenever its score, if the true test pair were (x, y), would be no larger than the $(1-\alpha)$ empirical quantile of the other scores. Because the scores are a symmetric function of the augmented sample, exchangeable data give exchangeable scores, and Lemma 1 yields Theorem 1: coverage of at least $1-\alpha$, and at most $1-\alpha+1/(n+1)$ without ties. The coverage is marginal over the training sample and the test point, not conditional on X_{n+1}.

3.2 Covariate shift and weighted conformal prediction

PAPER The data model is

$$\begin{aligned}(X_i, Y_i) &\overset{\text{iid}}{\sim} P = P_X \times P_{Y|X}, \quad i \le n,\\ (X_{n+1}, Y_{n+1}) &\sim \tilde{P} = \tilde{P}_X \times P_{Y|X}, \text{ independently.}\end{aligned} \tag{5}$$

Only the marginal law of the covariates changes. Let $w(x) = d\tilde{P}_X(x)/dP_X(x)$. The unweighted empirical law $\frac{1}{n+1}\sum_{i\le n}\delta_{V_i} + \frac{1}{n+1}\delta_\infty$ is replaced by

$$\begin{gathered}\sum_{i=1}^{n} p_i^w(x)\,\delta_{V_i(x,y)} + p_{n+1}^w(x)\,\delta_\infty,\\ p_i^w(x) = \frac{w(X_i)}{\sum_{j=1}^n w(X_j) + w(x)}, \quad p_{n+1}^w(x) = \frac{w(x)}{\sum_{j=1}^n w(X_j) + w(x)}.\end{gathered} \tag{6}$$

$w(x)$ Likelihood ratio of test to training covariate density; large where the test population is over-represented relative to training.

$p_i^w(x)$ Probability attached to training score i; proportional to how much its covariate resembles the test population.

$p_{n+1}^w(x)$ Mass at infinity, depending on the test covariate x itself through $w(x)$; it replaces the constant $1/(n+1)$.

The band of Corollary 1 is Equation (4) with the weighted law in place of the unweighted one:

$$\widehat{C}_n(x) = \left\{y :\ V_{n+1}(x,y) \le \text{Quantile}\left(1-\alpha,\ \sum_{i=1}^{n} p_i^w(x)\delta_{V_i(x,y)} + p_{n+1}^w(x)\delta_\infty\right)\right\}, \tag{7}$$

which satisfies $\mathbb{P}\{Y_{n+1} \in \widehat{C}_n(X_{n+1})\} \ge 1-\alpha$ under (5) when $\tilde{P}_X \ll P_X$. A constant factor in w cancels in (6), so the ratio is needed only up to proportionality (Remark 3).

INFERENCE Equation (6) produces an interval of infinite length at any x where $\sum_{i\le n} p_i^w(x) < 1-\alpha$. Writing $W = \sum_{j\le n} w(X_j)$, this occurs exactly when $W/(W+w(x)) < 1-\alpha$, that is when

$$w(x) \ > \ \frac{\alpha}{1-\alpha}\, W. \tag{8}$$

The paper does not state this condition, but it follows directly from (6) and it governs the behaviour of the method in the tails, as the simulation below shows.

3.3 Split conformal and the effective sample size

PAPER Refitting $\hat{\mu}$ for every candidate (x, y) is costly. Split conformal prediction fits μ_0 on a preliminary sample and uses the fixed score $\mathcal{S}((x,y),Z) = |y - \mu_0(x)|$, so that the weighted band becomes

$$\widehat{C}_n(x) = \mu_0(x) \pm \text{Quantile}\Big(1-\alpha,\ \sum_{i=1}^{n} p_i^w(x)\,\delta_{|Y_i-\mu_0(X_i)|} + p_{n+1}^w(x)\,\delta_\infty\Big), \tag{9}$$

valid conditionally on the preliminary sample. For the experiment the authors use an exponential tilt and the heuristic effective sample size:

$$w(x) = \exp(x^\top \beta),\ \ \beta = (-1,0,0,0,1); \qquad \hat{n} = \frac{\big[\sum_{i=1}^n |w(X_i)|\big]^2}{\sum_{i=1}^n |w(X_i)|^2} = \frac{\|w(X_{1:n})\|_1^2}{\|w(X_{1:n})\|_2^2}. \tag{10}$$

The ratio $\hat{n}$ equals n when all weights are equal and falls toward 1 when one weight dominates; it is the Kish approximation to the number of equally weighted observations that would carry the same variance. When the ratio is unknown, a classifier separates training points (class 0) from unlabeled test covariates (class 1), and Bayes' rule gives

$$\frac{\mathbb{P}(C=1 \mid X=x)}{\mathbb{P}(C=0 \mid X=x)} = \frac{\mathbb{P}(C=1)}{\mathbb{P}(C=0)}\,\frac{d\tilde{P}_X}{dP_X}(x), \qquad \hat{w}(x) = \frac{\hat{p}(x)}{1-\hat{p}(x)}. \tag{11}$$

The class prior is a constant and drops out by Remark 3.

3.4 Weighted exchangeability and the general theorem

PAPER The core of the paper is the conditioning argument. Condition on the unordered set of values, event $E_v : \{V_1,\dots,V_{n+1}\} = \{v_1,\dots,v_{n+1}\}$. With joint density f and exchangeability, for each i

$$\mathbb{P}\{V_{n+1} = v_i \mid E_v\} = \frac{\sum_{\sigma:\sigma(n+1)=i} f(v_{\sigma(1)},\dots,v_{\sigma(n+1)})}{\sum_{\sigma} f(v_{\sigma(1)},\dots,v_{\sigma(n+1)})} = \frac{n!}{(n+1)!} = \frac{1}{n+1}, \tag{12}$$

since all permuted densities are equal. The test score is therefore uniform over the observed multiset, which gives Lemma 1 again. Exchangeability enters only through this probability, which is what permits generalisation. **Definition 1.** Random variables $V_1,\dots,V_n$ are weighted exchangeable with weight functions $w_1,\dots,w_n$ if their density factorises as $f(v_1,\dots,v_n) = \prod_{i=1}^n w_i(v_i)\cdot g(v_1,\dots,v_n)$, with g invariant to permutation of its arguments. Independent draws $Z_i \sim P_i$, with $P_i \ll P_1$, are weighted exchangeable with $w_1 \equiv 1$ and $w_i = dP_i/dP_1$ (Lemma 2). In the weighted case the same computation

gives, because g is permutation invariant and cancels,

$$\mathbb{P}\{Z_{n+1} = z_i \mid E_z\} = p_i^w(z_1, \dots, z_{n+1}) = \frac{\sum_{\sigma:\sigma(n+1)=i} \prod_{j=1}^{n+1} w_j(z_{\sigma(j)})}{\sum_{\sigma} \prod_{j=1}^{n+1} w_j(z_{\sigma(j)})}. \tag{13}$$

Lemma 3 then states that for weighted exchangeable data $\mathbb{P}\{V_{n+1} \le \text{Quantile}(\beta; \sum_{i\le n} p_i^w \delta_{V_i} + p_{n+1}^w \delta_\infty)\} \ge \beta$, and Theorem 2 gives the corresponding band with the same coverage. For covariate shift, $w_i \equiv 1$ for $i \le n$ and $w_{n+1}(x, y) = w(x)$, so the sums in (13) collapse to $w(x_i)/\sum_{j\le n+1} w(x_j)$, which is Equation (6). This is the proof of Corollary 1.

CRITIQUE Remark 5 correctly notes that no useful upper bound exists in the weighted case, because the largest jump of the conditional distribution of V_{n+1} is $\max_i p_i^w$, which can be large, whereas it is always $1/(n+1)$ without weights. Weighted conformal bands are thus guaranteed not to undercover, but may overcover without limit; the paper's experiment shows modest overcoverage (90.8 percent against 90 percent) while the simulation below shows more at strong shifts.

3.5 Local conditional coverage

PAPER Exact conditional coverage $\mathbb{P}\{Y_{n+1} \in \hat{C}_n(x_0) \mid X_{n+1} = x_0\} \ge 1 - \alpha$ is impossible without infinite expected length (Vovk, 2012; Lei and Wasserman, 2014). The authors relax it to a kernel-smoothed average,

$$\frac{\int \mathbb{P}\big(Y_{n+1} \in \hat{C}_n(x_0) \mid X_{n+1} = x\big) K\big(\tfrac{x - x_0}{h}\big) dP_X(x)}{\int K\big(\tfrac{x - x_0}{h}\big) dP_X(x)} \;\ge\; 1 - \alpha, \tag{14}$$

which is a covariate-shift problem with $d\tilde{P}_X/dP_X \propto K((\cdot - x_0)/h)$. Weighted conformal prediction with weights $K((X_i - x_0)/h)$ delivers it, but the band depends on the centre x_0 and must be recomputed for each x_0, so no single band with this property at all centres is produced.

4 Data and Empirical Strategy

4.1 Data

PAPER The experiment uses the UCI airfoil self-noise dataset, $N = 1503$ observations of a response Y (scaled sound pressure level) and $d = 5$ covariates (log frequency, angle of attack, chord length, free-stream velocity, suction side log displacement thickness). Nothing is financial or time-indexed. For each of 5000 trials the data are randomly partitioned into D_{pre} (25 percent, used to fit μ_0 by linear regression), D_{train} (25 percent, the calibration sample), and D_{test} (50 percent, exchangeable with training). A shifted test set D_{shift} is formed by resampling 25 percent of D_{test} with replacement with probability proportional to $\exp(x^\top \beta)$, $\beta = (-1, 0, 0, 0, 1)$.

Table 1. **Experimental design reported in the paper**

Set	Role	Share of data	Approx. size
D_{pre}	Prefit linear μ_0	25 percent	376
D_{train}	Calibration scores	25 percent	376
D_{test}	Test, no shift	50 percent	751
D_{shift}	Test, exponential tilt	25 percent of D_{test}, resampled	188

Source: proportions from the paper (official); counts computed by the reviewer as rounded shares of $N = 1503$ (generated).

4.2 Empirical specification and identification

EVIDENCE The identifying variation is controlled by the experimenter: the covariate distribution of the test set is moved by a known exponential tilt while $P_{Y|X}$ is untouched because resampling whole observations preserves the conditional law. This is a clean design for testing the coverage theorem, because the covariate shift assumption holds by construction and the oracle weight is exactly known. It is not an identification strategy for a real shift, where the assumption $\tilde{P}_{Y|X} = P_{Y|X}$ is the substantive and untestable part.

4.3 Data limitations and biases

- Resampling the test set with replacement from D_{test} duplicates observations, so the shifted set contains repeated points and has strong within-set dependence; the reported coverage histograms average over test points but the effective test sample is small.
- The shifted population is a reweighting of the same finite dataset, so the shift cannot be larger than the support of that dataset allows; the heavy-tail behaviour of Equation (8) is not probed.
- A single, low-dimensional, tabular engineering dataset with a log-linear shift favours logistic regression as a weight estimator; the experiment cannot reveal behaviour in high dimension or under misspecified weights.
- No financial, time-series or dependent data are examined, so serial dependence, which violates weighted exchangeability, is outside the evidence.

5 Results

5.1 Loss of coverage of ordinary split conformal

EVIDENCE At nominal level 90 percent, ordinary split conformal averages 90.2 percent coverage over 5000 trials on D_{test} but only 82.2 percent on D_{shift}. The loss is 7.8 percentage points: nearly one in five shifted test points falls outside an interval that should miss one in ten. The mechanism is heteroskedasticity. The top row of Figure 1 in the paper shows that noise level varies with the first and fifth covariates, which are exactly those the tilt up-weights, so the calibration quantile, learned on the training mixture, understates the residual spread in the tilted region. The lengths of ordinary intervals are the same with or without shift, as the authors note, since only the test population changes.

5.2 Oracle weights and the effective sample size

EVIDENCE Weighted split conformal with the true weight (10) brings average coverage to 90.8 percent, a 0.8 point overshoot, but the coverage histogram is more dispersed. The authors explain this by the effective sample size $\hat{n}$ and verify it by running ordinary conformal with only $\hat{n}$ subsampled training points; the resulting histogram nearly coincides with that of weighted conformal. This is a strong and useful check: it shows that the extra dispersion is entirely the cost of information loss from unequal weights rather than a defect of the method. Weighted intervals are also longer than the unweighted interval at equal effective sample size, which the authors attribute to μ_0 not being adapted to the shifted distribution.

5.3 Estimated weights

EVIDENCE Logistic regression and random forest class-probability estimates, inserted in (11), give average coverage of 91.0 percent each. Logistic regression is correctly specified because the log-odds are linear in x. Random forests, more flexible, produce slightly more dispersed and in some cases much longer intervals. Without shift, weighting changes little for logistic regression and slightly increases dispersion for random forests (Figure 4 of the paper).

5.4 Reviewer's reconstruction: a controlled simulation

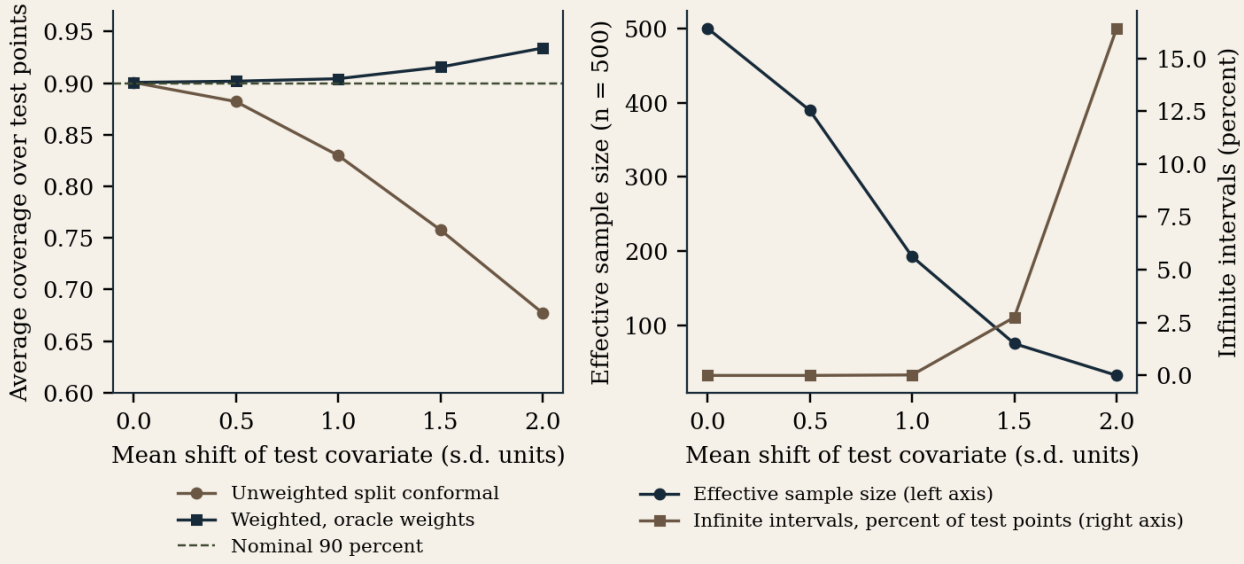
EVIDENCE Because the airfoil figures could not be inspected, the reviewer ran an independent simulation to examine the behaviour across shift magnitudes. The design is one-dimensional: $X \sim N(0,1)$ on training, $Y = X + (0.3 + 0.5|X|)\varepsilon$ with $\varepsilon \sim N(0,1)$, a linear μ_0 fitted on a separate sample of 500, calibration size $n = 500$, test points $X \sim N(\mu, 1)$ so that $w(x) = \exp(\mu x - \mu^2/2)$ exactly, $\alpha = 0.1$, and 1000 repetitions with 2000 test points each. Results are generated by the reviewer, not taken from the paper.

Table 2. **Simulation of weighted and unweighted split conformal under Gaussian mean shift (generated by the reviewer)**

Shift μ	Unweighted coverage	Weighted coverage	S.d., weighted	Effective $\hat{n}$	Infinite	Length ratio
0.0	0.901	0.901	0.015	500	0.0 pct	1.00
0.5	0.882	0.902	0.019	390	0.0 pct	1.08
1.0	0.830	0.904	0.032	193	0.0 pct	1.31
1.5	0.758	0.916	0.044	76	2.8 pct	1.62
2.0	0.677	0.934	0.047	33	16.4 pct	1.87

Source: generated by the reviewer (seed 20261021, 1000 repetitions, 500 calibration points, 2000 test points). Pct is percent of test points; length ratio is median weighted over unweighted length, computed on finite intervals only.

Figure 1. **Coverage, effective sample size and infinite intervals as the covariate mean shift grows**



Source: generated by the reviewer; Gaussian mean-shift simulation described in the text, 1000 repetitions, nominal level 90 percent, calibration size 500.

The results confirm and extend the paper. Unweighted coverage falls monotonically from 90 percent to 68 percent as the shift grows to two standard deviations, whereas weighted coverage stays at 90 to 93 percent. The effective sample size collapses from 500 to 33. With Gaussian tilt, $\mathbb{E}_P[w^2] = e^{\mu^2}$, so the population analogue of $\hat{n}$ is $ne^{-\mu^2}$: 184 at $\mu = 1$, against 193 simulated, and 9 at $\mu = 2$, against 33 simulated. The simulated value is higher at $\mu = 2$ because the finite-sample maximum weight truncates the heavy tail of w, so the population formula overstates the loss. Finally, at $\mu = 2$ 16.4 percent of test points receive an infinite interval through Equation (8), which makes the weighted overcoverage (93.4 percent) partly an artifact of trivially valid intervals.

6 Robustness and Additional Tests

Table 3. **Robustness: concern, test, result and assessment**

Concern	Test reported	Result	Does it resolve the concern?
Dispersion of weighted coverage may reveal a flaw	Unweighted conformal using only $\hat{n}$ training points	Histogram nearly coincides with weighted conformal	Yes for dispersion; it shows information loss, not error
Weights unknown in practice	Logistic regression and random forest density ratio estimates	Coverage 91.0 percent each	Partly: one low-dimensional well-specified example, no theory
Weighting may harm when there is no shift	Weighted conformal on D_{test} with estimated weights	Logistic similar to unweighted, forest slightly more dispersed	Partly: shows no large harm in a small, low-dimensional case
Result may depend on one split	5000 random resplits	Averages and histograms reported	Yes for split variability; the dataset itself is unchanged
Intervals may be uselessly wide	Median length histograms	Weighted longer than unweighted at equal $\hat{n}$	Partly: lengths reported, but no benchmark against alternative methods

Source: tests and results as reported in the paper (official); assessment by the reviewer.

CRITIQUE The paper reports no placebo test of the covariate shift assumption, no variation of the shift strength, no alternative datasets and no comparison with competing procedures such as covariate-shift-adjusted regression bands. The reviewer's simulation in Table 2 supplies the shift-strength variation.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

Under weighted exchangeability, and in particular under covariate shift with a known likelihood ratio and independent draws, the weighted conformal band (7) has marginal coverage of at least $1 - \alpha$ in finite samples. This is a theorem with a complete proof, and the reviewer's simulation agrees with it.

SUPPORTED BUT CONDITIONAL

Estimated weights give approximately nominal coverage, but only in a low-dimensional example in which the classifier is correctly specified and the oracle weights are well behaved, and without a theorem.

SUGGESTIVE

The graphical-model and missing-covariate applications are plausible reductions to a lower-dimensional likelihood ratio, but they are stated without proofs or experiments.

NOT ESTABLISHED

Conditional coverage given X_{n+1}, robustness to a misestimated ratio, performance in high dimension, validity under serial dependence, and any statement about whether the covariate shift assumption holds in a given application.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- The method is as cheap as ordinary conformal prediction under covariate shift, and works with any base algorithm and any score, which matters for adoption.
- Weighted exchangeability is introduced as a general concept that includes independent non-identically distributed data, not only the shift case.
- The reduction of the effective sample size is quantified and tied to the dispersion seen in the experiment.

STRENGTHS IDENTIFIED INDEPENDENTLY

- The proof is short and transparent: exchangeability is isolated as the source of one uniform probability statement, so the generalisation requires changing a single computation. Later work in the literature (Barber et al., 2023) builds on this structure.
- Placing the test point's own weight in the denominator and at infinity keeps the guarantee exact rather than asymptotic; a quantile of the reweighted training scores alone, without the test point, would not carry the guarantee.
- The ESS-matching control experiment is a good use of a falsifiable prediction: if dispersion is an ESS effect, matching ESS should remove it, and it does.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. The guarantee is conditional on an oracle likelihood ratio, but the paper offers neither a theorem nor a diagnostic for the estimated ratio that practice requires. **Why it matters.** In realistic settings with many covariates the ratio is estimated, may be heavy-tailed, and is not identifiable where the training support is thin; since p^{w}_{n+1} enters the quantile directly, errors in $\hat{w}$ at the test point shift the quantile level itself. **Potential bias.** Undercoverage whenever $\hat{w}$ underestimates the true ratio in regions where test mass is high; overcoverage and infinite intervals when it overestimates. **How**

it could be tested. Contaminate the oracle ratio by controlled multiplicative noise or misspecified classifiers in simulations of the type in Table 2, and measure coverage as a function of the error in $\log \hat{w}$; later literature gives bounds in terms of total variation between the true and estimated weights (Barber et al., 2023).

OTHER WEAKNESSES

- Marginal rather than conditional coverage: the guarantee averages over test covariates, so some subpopulations may be badly under-covered even when average coverage is nominal. Test by stratifying coverage on $w(X)$ or on covariate bins.
- Reliance on the covariate shift assumption $\tilde{P}_{Y|X} = P_{Y|X}$, which cannot be verified from labeled training data and unlabeled test covariates alone. A shift in the conditional law, such as a regime change in return dynamics, invalidates the guarantee without any warning from the method.
- Independence of draws is assumed. Time series and clustered data are not weighted exchangeable, and the paper does not caution adequately about this.
- Experimental evidence is thin: one dataset, one shift family, one base learner, and lengths are not compared with competing approaches.
- The weighted variant, based on an arbitrary score, is not guaranteed to be efficient; the intervals can become infinite in the tails without an indication in the paper, as Equation (8) shows.

8 Contradictions and Edge Cases

CRITIQUE The reviewer examined when the guarantee should fail or become uninformative.

First, the absolute continuity requirement $\tilde{P}_X \ll P_X$ is necessary. If the test population puts mass where training has none, w is undefined and no weighting can help; the result is unavoidable extrapolation. Second, the ratio is only required up to scale, but the shape must be right, because the quantile level depends on the relative size of $w(x)$ and W.

Third, heavy tails. Let $w(x)$ grow with x. For a Gaussian mean shift, $w(x) = \exp(\mu x - \mu^2/2)$ and $\mathbb{E}_P[w] = 1$, but the second moment is e^{μ^2} and the population effective sample size is $ne^{-\mu^2}$. At $\mu = 3$ this is below 0.1 observations out of 500. The method remains formally valid yet becomes uninformative, since the infinite-interval condition (8) then applies to most test points. In the simulation of Table 2, 16.4 percent of test points had infinite intervals at $\mu = 2$ and the effective sample size was 33; for larger μ the share rises quickly.

Fourth, the overcoverage is unbounded. Remark 5 shows that the upper bound fails; in the simulation, weighted coverage reaches 93.4 percent, partly through infinite intervals. A user who sees nominal or higher coverage would not detect that the intervals are trivial at the tail.

Fifth, a covariate shift in a variable that is a deterministic function of others. If the training sample contains an exact function relationship, the density ratio may be undefined or numerically unstable.

Sixth, an alternative mechanism can generate the same observed undercoverage as covariate shift: a shift in the conditional variance with unchanged covariates. Weighted conformal prediction would repair the first but be silent about the second; coverage would still fail with no signal. A real example is a change in volatility regime for the same observable state variables.

Finally, serial dependence. With positively autocorrelated residuals, the effective number of independent calibration scores is smaller than n, and the exchangeability-based rank statement does not hold. Combined with a covariate shift, which time series commonly exhibit, neither the weighting nor the standard theory applies.

9 Further Literature

Table 4. **Literature connections**

Paper	Shared idea	Relationship	Why it matters
Vovk, Gammerman and Shafer (2005)	Conformal prediction under exchangeability	Foundation; this paper extends its guarantee beyond exchangeability	Supplies the construction and the rank argument that the weighted lemma generalises
Lei, G'Sell, Rinaldo, Tibshirani and Wasserman (2018)	Distribution-free predictive inference for regression, upper bound	Preceded and is used directly	Provides the regression implementation and the near-tight upper bound that the weighted case lacks
Shimodaira (2000)	Importance weighting of the likelihood under covariate shift	Developed the same weighting idea for estimation	Shows that the ratio is the central object in covariate shift, here transferred to intervals
Sugiyama, Suzuki and Kanamori (2012)	Density ratio estimation	Supplies the estimation toolbox for the ratio	Covers the probabilistic classification approach used in Equation (11)
Vovk (2012); Lei and Wasserman (2014)	Impossibility of conditional coverage	Explain why the paper settles for marginal and locally smoothed coverage	Mark the boundary of what weighted conformal can offer
Barber, Candès, Ramdas and Tibshirani (2021)	Limits of distribution-free conditional inference	Companion paper by the same authors	Formalises what cannot be achieved, which motivates Equation (14)
Gibbs and Candès (2021)	Adaptive conformal inference under distribution shift	Different approach to shift: online adjustment of the level	Handles arbitrary shift in time series without a known ratio, trading guarantees for long-run coverage
Barber, Candès, Ramdas and Tibshirani (2023)	Conformal prediction beyond exchangeability	Extended and quantified the weighting idea	Gives coverage gaps in terms of the total variation distance and handles dependence and estimated weights
Podkopaev and Ramdas (2021)	Conformal classification under label shift	Applied the weighted idea to label shift	Shows that weighting by a different ratio fits other shift types
Lei and Candès (2021)	Conformal inference of counterfactuals and treatment effects	Applied the weighted construction to causal inference	Propensity-score ratios play the role of w, an economically relevant use case

Source: reviewer's classification; relationships are described from the cited papers' known subject matter and were not verified by reading the full text of each.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** No guarantee for estimated weights. *Why it matters:* It is the case that arises in practice. *Existing limitation:* Only simulation in one example. *Way forward:* Bound the coverage gap by the discrepancy between $\hat{w}$ and w, as in later work, and design diagnostics.
- **Gap:** No treatment of dependent data. *Why it matters:* Financial and macroeconomic data are serially dependent. *Existing limitation:* Independence is built into Lemma 2. *Way forward:* Block or mixing-based weights, or online recalibration.

10.2 Literature-level gaps

- **Gap:** Conditional coverage under shift. *Why it matters:* Decisions concern subpopulations. *Existing limitation:* Impossibility results for exact conditional coverage. *Way forward:* Group-conditional or locally weighted guarantees with explicit assumptions.
- **Gap:** Testing the covariate shift assumption. *Why it matters:* The guarantee fails silently otherwise. *Existing limitation:* The assumption concerns an unobserved conditional law on the test population. *Way forward:* Sensitivity analyses that parametrise a bounded departure from $\tilde{P}_{Y|X} = P_{Y|X}$.

10.3 Methodological gaps

- **Gap:** Stabilisation of heavy-tailed weights. *Why it matters:* The infinite-interval condition (8) makes strong shifts uninformative. *Existing limitation:* Clipping is not covered by the theorem. *Way forward:* Characterise the coverage price of clipping at level c in terms of the tail mass of w.
- **Gap:** Computation of the full weighted conformal band under general weighted exchangeability. *Why it matters:* The sum in (13) is over $(n+1)!$ permutations. *Existing limitation:* Only the covariate shift case collapses to a closed form. *Way forward:* Sampling or dynamic-programming approximations with controlled error.

11 Research Opportunities

Idea 1. Weighted conformal return forecasts across volatility regimes

RESEARCH QUESTION. Do weighted conformal intervals for next-day equity returns keep nominal coverage when the test period has a different distribution of state variables (VIX, term spread) than the training period?

MOTIVATION. Return forecasts are trained on calm periods and used in turbulent ones.

HYPOTHESIS. Weighting by the density ratio of state variables restores coverage in stress months, but dispersion and infinite intervals rise sharply.

ECONOMIC MECHANISM. Conditional volatility depends on observable state variables; a shift in their distribution is a covariate shift if the conditional return law given the state is stable.

LITERATURE CONNECTION. Tibshirani et al. (2019); Gibbs and Candès (2021); Barber et al. (2023).

DATA. Daily returns of a broad index and Fama-French factors from the Kenneth French data library; VIX and Treasury yields from FRED. Availability to be verified before the project.

VARIABLES. Next-day return, lagged returns, VIX level, term spread, realised volatility.

METHOD. Split conformal with a linear or gradient boosting base model; ratio estimated by a classifier between training and test windows; rolling-window evaluation.

IDENTIFICATION STRATEGY. Evaluate empirical coverage out of sample in pre-specified stress windows and stratify by estimated weight.

EXPECTED CONTRIBUTION. An empirical test of whether the shift assumption is plausible for financial state variables.

MAIN RISK. Serial dependence and conditional-law shifts violate the assumptions, so the experiment may mostly document failure.

Idea 2. Size-segment transfer in machine learning asset pricing

RESEARCH QUESTION. Can prediction intervals for stock returns trained on large capitalisation firms be transferred to small capitalisation firms through covariate weighting?

MOTIVATION. Machine learning return models are usually fitted on pooled data dominated by large stocks.

HYPOTHESIS. Characteristic distributions differ strongly across segments, and weighting by the ratio of characteristics corrects coverage only partly because the conditional return law differs.

ECONOMIC MECHANISM. Limits to arbitrage and illiquidity make the conditional return law differ in small stocks, violating the shift assumption in a measurable way.

LITERATURE CONNECTION. Tibshirani et al. (2019); Gu, Kelly and Xiu (2020) is the prediction model of the library and a natural base learner.

DATA. CRSP and Compustat characteristics, or a public substitute such as the open characteristics panel; availability and licensing to be verified.

VARIABLES. Monthly returns, size, book-to-market, momentum, volatility, turnover.

METHOD. Weighted split conformal with classifier-based ratios between segments; compare with segment-specific recalibration.

IDENTIFICATION STRATEGY. The gap in coverage between weighted transfer and the segment-specific benchmark measures the size of the conditional-law shift.

EXPECTED CONTRIBUTION. A quantitative decomposition of the cross-segment shift into covariate and conditional components.

MAIN RISK. Data access and weak overlap in characteristics between segments.

Idea 3. Effective sample size and weight clipping: a finite-sample theory

RESEARCH QUESTION. What coverage and length does clipped weighted conformal achieve as a function of the clipping level and the tail of the ratio?

MOTIVATION. Equation (8) and the simulation show that strong shifts yield infinite or very wide intervals.

HYPOTHESIS. Clipping weights at c loses at most a coverage amount proportional to the test mass where $w > c$, while reducing the infinite-interval fraction.

ECONOMIC MECHANISM. Not applicable; a statistical trade-off between bias and variance.

LITERATURE CONNECTION. Tibshirani et al. (2019); Barber et al. (2023); Sugiyama et al. (2012).

DATA. Simulation first, in the design of Table 2 with Gaussian, Student and exponential tilts; then a public tabular benchmark such as the UCI datasets.

VARIABLES. Clipping level, shift strength, tail index, calibration size.

METHOD. Derive coverage bounds with a total-variation term, verify by simulation.

IDENTIFICATION STRATEGY. Controlled simulation with known ratios.

EXPECTED CONTRIBUTION. A practical rule for the maximum tolerable shift for a given sample size.

MAIN RISK. Overlap with the existing bounds in the later literature.

Idea 4. Scenario-conditional loss intervals for bank stress tests

RESEARCH QUESTION. Can conformal intervals for loan loss rates trained on historical conditions be reweighted to a regulatory adverse scenario?

MOTIVATION. Stress scenarios are by definition outside the typical training distribution of macroeconomic drivers.

HYPOTHESIS. Coverage under the adverse scenario degrades because the scenario lies in the thin tail of the training distribution, producing very small effective sample sizes.

ECONOMIC MECHANISM. Credit losses depend on unemployment and house prices; a scenario shifts their distribution, while the loss function given the drivers is assumed stable.

LITERATURE CONNECTION. Tibshirani et al. (2019); Lei and Candès (2021).

DATA. Net charge-off rates from FFIEC call reports or FRED, and published supervisory scenario paths from the Federal Reserve; availability to be verified.

VARIABLES. Charge-off rate, unemployment, house price index, GDP growth.

METHOD. Weighted split conformal using the scenario distribution as test covariate law, with density ratio from a parametric fit to historical drivers.

IDENTIFICATION STRATEGY. Back-test on the 2008 to 2010 period, using training data before 2007.

EXPECTED CONTRIBUTION. A transparent measure of how much the scenario exceeds what history can support.

MAIN RISK. Short macroeconomic samples and a structural break in the conditional law in crises.

Idea 5. Dependence-robust weighted conformal for time series

RESEARCH QUESTION. How much does serial dependence reduce the coverage of weighted conformal prediction, and can blocking restore it?

MOTIVATION. All the paper's guarantees assume independent draws.

HYPOTHESIS. Coverage deficiency grows with the autocorrelation of scores, and a block-based calibration recovers most of it.

ECONOMIC MECHANISM. Volatility clustering makes absolute residuals positively autocorrelated.

LITERATURE CONNECTION. Tibshirani et al. (2019); Xu and Xie (2021); Gibbs and Candès (2021).

DATA. Simulated GARCH and autoregressive processes, then daily returns.

VARIABLES. Score autocorrelation, block length, shift magnitude.

METHOD. Monte Carlo coverage comparison across calibration schemes.

IDENTIFICATION STRATEGY. Known data-generating process in simulation.

EXPECTED CONTRIBUTION. Quantified limits of the theory for financial time series.

MAIN RISK. Results may reproduce what the later literature has already shown.

11.1 Ranking

Table 5. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Volatility regimes	3	5	5	4	4	3	24
2. Size-segment transfer	4	3	3	4	4	4	22
3. Clipping theory	3	4	5	4	3	3	22
4. Stress-test scenarios	4	3	3	3	5	4	22
5. Dependence-robust	2	4	5	4	4	2	21

Source: reviewer's subjective scoring.

The top three by total are Ideas 1, 2 and 4 (the last two tied with Idea 3 at 22; the tie is broken in favour of the ideas with greater economic importance and novelty). Idea 1 is the most feasible and gives direct evidence on whether the shift assumption is credible in markets. Idea 2 decomposes a known cross-sectional difference into covariate and conditional components, which is of independent economic interest. Idea 4 addresses a policy-relevant setting where extrapolation beyond history is the point of the exercise.

12 Learning Framework

12.1 Prerequisite concepts

- Exchangeability and its relation to i.i.d. sampling; order statistics and ranks.
- Quantiles of discrete distributions and the generalised inverse.
- Radon-Nikodym derivatives and likelihood ratios; importance sampling.
- Regression residuals, heteroskedasticity, and the difference between marginal and conditional coverage.

12.2 Core concepts learned

- Conformal prediction converts any predictor into an interval with finite-sample coverage through the rank of one score among $n + 1$.
- Weighted exchangeability factorises the joint density into a permutation-invariant part and a product of one-point weights.
- The test point's weight must enter the normalisation and the mass at infinity.
- Effective sample size summarises the information lost to weighting.

12.3 Important equations

Equation (1) is the exchangeable quantile lemma. Equation (6) gives the weights and Equation (7) the weighted band. Equation (13) is the central probability of weighted exchangeability. Equation (10) gives the effective sample size, and Equation (11) the classifier-based ratio. Equation (8), derived in this review, gives the infinite-interval condition.

12.4 Methods learned

Split conformal calibration, weighted quantiles with a mass at infinity, density ratio estimation by probabilistic classification, and coverage evaluation by repeated random splitting.

12.5 Questions to test understanding

COMPREHENSION

1. Why does conformal prediction need the mass at infinity, and what coverage is lost if it is omitted?
2. What changes in the weights when the test covariate is far in the tail of the training distribution?
3. What does it mean that coverage is marginal and not conditional?

ANALYSIS

1. Prove Equation (13) reduces to Equation (6) when only the last weight function differs from one.
2. Show that the odds ratio in Equation (11) equals the likelihood ratio up to a constant.
3. Derive the condition for an infinite interval and the population value of the effective sample size under a Gaussian mean shift.

RESEARCH

1. How would the guarantee degrade if the conditional law shifts by a bounded total-variation amount?
2. Can weights be chosen to minimise interval length subject to coverage?
3. What is the right notion of weighted exchangeability for a stationary time series?

12.6 Papers to read next

Vovk, Gammerman and Shafer (2005); Lei et al. (2018); Barber et al. (2023); Gibbs and Candès (2021); Romano, Patterson and Candès (2019); Angelopoulos and Bates (2021).

13 Conclusion

13.1 One-sentence thesis

Conformal prediction survives covariate shift if each calibration score is weighted by the test-to-training likelihood ratio and the test point's own weight is placed at infinity, because coverage depends only on a weighted-exchangeability computation.

13.2 Three most important insights

- Exchangeability is used in conformal prediction only to compute the probability that the test score takes each observed value; replacing that computation with the weighted version generalises the theory.
- The price of correction is a reduced effective sample size, which makes intervals wider and coverage more variable, and which can make them infinite in the tail.
- The guarantee is marginal, assumes independence, assumes the conditional law is unchanged and requires the ratio.

13.3 Strongest evidence

The theorem with its proof, together with the effective-sample-size control experiment in which unweighted conformal with $\hat{n}$ subsampled points reproduces the dispersion of the weighted method.

13.4 Biggest weakness

No guarantee or diagnostic for estimated weights, which is what practice requires.

13.5 Most important research gap

Quantifying the coverage loss from errors in the estimated ratio and from violations of the covariate shift assumption.

13.6 Best research opportunity

Testing weighted conformal prediction on financial state variables across volatility regimes, where the plausibility of the shift assumption is itself the finding.

13.7 Three papers to read next

Barber et al. (2023); Gibbs and Candès (2021); Lei and Candès (2021).

14 References

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